ZOO302 Current Solved

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Q.1 Explain Stomach in Ruminants? Long

• Ruminant mammals are the animals that “chew their cud,’’ such as cows, sheep, and deer show some of the most unusual modifications of the stomach.

• This method of digestion has evolved in animals that need to eat large amounts of food relatively quickly, but can chew the food at a more comfortable or safer location.

• The ruminant stomach provides an opportunity for large numbers of microorganisms to digest the cellulase walls of grass and other vegetation.

• Cellulose contains a large amount of energy; however, animals generally lack the ability to produce the enzyme cellulose for digesting cellulose and obtaining its energy.

• In ruminants, the upper portion of the stomach expands to form a large pouch, the rumen, and a smaller reticulum.

Q.2 Digestion in grasshopper?

• In grasshopper (or cockroach) the digestion of food starts in the mouth. The saliva contains some enzymes which digest starch present in the food. Saliva also helps in lubrication and softening of food. This partially digested food passes through the esophagus into crop where it is stored temporarily. Then, the food passes into gizzard. Here, the food is finely ground by tough walls of gizzard. From gizzard, the food goes into stomach. In the stomach, the enzymes secreted by elongated glands called hepatic caeca get mixed with the food to digest it completely.

Q.3 Osmoregulation defines?

• **It is defined as** the maintenance of constant osmotic pressure in the fluids of an organism by the control of water and salt concentrations.

• It means osmoregulation governs water and salt balance, and excretion of metabolic wastes.

Q.4 what is digestion?

Digestion is the chemical and/or mechanical breakdown of complex food into diffusible molecules i.e. particles that individual cells of an animal can absorb.

• Starch into glucose.

• Proteins into amino acids.

• Fats into fatty acids & glycerol.

Q.5 what is Predation?

Predation is one of the most sophisticated feeding strategies, since it requires the capture of live prey.

It is an interaction in which one organism kills another for food. The organisms that does the killing for food is the predator and the organisms that is killed for food is the prey.

Examples:

Wolves hunting a deer or moose, an owl hunting the mice

Q.6 Intracellular and Extracellular digestion? Long

Intracellular digestion	**Extracellular digestion**
In intracellular digestion, the breakdown of food materials into small molecules occurs inside the food vacuoles within the cell.	In extracellular digestion, the breakdown of food materials into small molecules occurs outside the cell in the lumen of the alimentary canal or on the decaying organic materials.
In intracellular digestion breakdown	In extra cellular digestion, the

of food occurs within the cells. First food in taken into the cell vacuole The cytoplasm secretes digestive enzymes into the food vacuole. Enzymes break larger molecules into smaller ones, which enter the cytoplasm.	enzymes are secreted outside the cell into the gut cavity or lumen where digestion occurs. Nutrients from the food then pass into body cells for energy metabolism or biosynthesis

Q.7 Gallbladder function?

• The gallbladder is a small bag near the liver.

• The gallbladder stores the greenish fluid.

Gallbladder is a small hollow organ where bile is stored and concentrated before it is released into the small intestine.

• The structure and position of the gallbladder can vary significantly among animal species. Because of the importance of bile in fat digestion, the gallbladder is relatively large in carnivores and vertebrates.

Q.8 Temperature Regulations in Fishes? **Long**

How Fishes live in extremely cold water?

1. They have “antifreeze” materials in their blood in the form of Polyalcohols (e.g., sorbitol, glycerol) or water soluble peptides and glycopeptides.

• Because these substances lower the freezing point of blood plasma and other body fluids.

2. These fishes also have proteins or protein-sugar compounds that inhibit the growth of ice crystals that begin to form.

3. These adaptations enable these fishes to stay flexible and swim freely in a supercooled state (i.e., at a temperature below the normal freezing temperature of a surrounding water).

For example: Bluefin tuna and the great white shark.

Q.9 Functions of Liver in vertebrates?

• Bilirubin (“red bile’’) and biliverdin (“green bile’’).
• Liver detoxifies various metabolites.
• Liver synthesizes proteins and produces bio chemicals necessary for digestion and growth.
• Destruction of RBCs and formation of bile pigments.
• Protein metabolism
• Synthesis of plasma proteins
• Enzyme synthesis.
• Conversion of ammonia to urea.

Q.10 Antiperistalsis?

• The reverse peristalsis is called Antiperistalsis, which occurs occasionally.
• These movements cause the passage of food from the intestine back into the stomach or even into the mouth. This is also called vomiting.
• During peristalsis the food advances through the GI tract when the rings of circular smooth muscle contract behind it and relax in front of it.

Q.11 what is Gastrointestinal Motility? GI long

• As with any organ, the function of the gastro intestinal tract is determined by the type of tissues it contains.
• Most of the mammalian gastro intestinal tract has the same anatomical structure along its entire length.
• From the outside inward is a thin layer of connective tissue called the **serosa.**
• The serosa forms a moist epithelial sheet called the **peritoneum.**
• This peritoneum lines the entire abdominal cavity and covers all internal organs.
• Next are the longitudinal smooth-muscle layer and circular smooth-muscle layer.
• Underneath this muscle layer is the submucosa.
• The submucosa contains connective tissue, blood, and lymphatic vessels.

Q.12 Temperature Regulation in Mammals? Long

• Mammals like birds are also active and behaviorally complex vertebrates.

• They can live in habitats all over the earth because they are homeothermic endotherms.
• They can maintain body temperatures between 35 and 42° C with metabolic heat.
• Mammals that live in cold regions, such as the arctic fox and barren-ground caribou, also have these exchange vessels in their extremities (e.g., legs, tails, ears, and nose).
• Animals in hot climates, such as jackrabbits, have mechanisms (e.g., large ears) to get rid of excess heat from the body surface.

Q.13 Write nutrition and its types?

Nutrition includes all of those processes by which an animal takes in, digests, absorbs, stores, and uses food (nutrients) to meet its metabolic needs.
A nutrient is a component in foods that an organism uses to survive and grow.

Q.14 Small Intestine?

• The human small intestine is about 4 cm in diameter and 7 to 8 m in length. The small intestine (small bowel) is about 20 feet long and about an inch in diameter.
• Length of human small intestine is intermediate between the small intestines of typical carnivores and herbivores of similar size.
• It reflects the human's omnivorous eating habits.

Q.15 Describe primitive strike of birds. 3m

How and when primitive streak develop in the birds?

The major structural characteristic of avian, gastrulation is the primitive streak.

• This streak is first visible as a thickening of the epiblast at the posterior region of the embryo.The streak eventually extends 60 - 75% of the length of the area pellucida. • The primitive streak defines the axes of the embryo.

Q.16 How teleosts fishes osmoregulate in fresh and marine water?

Long

There seminal study demonstrated that freshwater teleosts extracts NaCl from the environment but marine water teleosts ingests seawater absorb intestinal water by absorbing NaCl. A marine fish has an internal osmotic concentration lower than that of the surrounding seawater, so it tends to lose water and gain salt.

Freshwater teleosts are hyperosmotic to the surrounding solution. Freshwater fish have very efficient kidneys that excrete water quickly. The internal environment of marine fish is hypo osmotic, while that of a Freshwater fish is hyperosmotic.